

PSID-SHELF · 1968-2021

PSID Studio

Setup Guide

Version 6.0 · June 2026

34 Topics

6,585 Variables

84,121 Individuals

A plain-English walkthrough for browsing PSID-SHELF online, or installing PSID Studio locally with your own full-resolution data.

What is PSID Studio?

PSID Studio is a web app that lets you **search, browse, and preview data** from the PSID-SHELF dataset — 34 topics of harmonized longitudinal data covering 1968–2021. You do not need to know any programming to use it.

Start here — which version do you need?

✓ Track A — Just browse online (no install needed)

Go to: psid-shelf-app.netlify.app — the app opens in your browser immediately. 34 topics, 10-row preview per topic. **If this is enough for your needs, stop here. You are done.**

↓ Track B — Full local install with your own data

Choose this if you want 100 rows per topic and offline access. Takes 30–60 minutes, mostly waiting for a large file to download. You will need to type two commands in a terminal — this guide walks you through every step.

At a glance

	PSID Studio Lite (online demo)	PSID Studio (full local version)
Topics	34	34
Data shown	10 rows per topic (baked in)	100 rows per topic (your data)
Where it runs	In your browser, online	On your computer, locally
Data download needed?	No	Yes — SHELF V2.1 from ICPSR
Badge shown	PREVIEW · 10-ROW SAMPLE	YOUR DATA · 100-ROW PREVIEW

Before you start (Track B only)

Check 1 — RAM (memory)

Your computer needs at least **8 GB of RAM**. The data preparation step loads a 2.2 GB file into memory. If you have less than 8 GB, it will fail silently mid-run with no useful error message.

- **Mac:** Apple menu → About This Mac → look for Memory.
- **Windows:** Settings → System → About → look for Installed RAM.
- **Linux:** open a terminal and type `free -h`

If you have less than 8 GB, run the data preparation step on a more powerful machine and copy the results over.

Check 2 – ICPSR account

You need a free account at openicpsr.org to download the SHELF data. Create one now if you do not have one.

PART 1

Get the app files

1. Go to: github.com/JF11579/psid-shelf-app

GitHub repo page — green Code button visible near top right

2. Click the green "< > Code" button near the top right.

Dropdown showing Download ZIP option

3. Click **Download ZIP**.
4. Find **psid-shelf-app-main.zip** in your Downloads folder and unzip it:
 - Mac: double-click the zip file.
 - Windows: right-click → Extract All.
 - Linux: right-click → Extract Here.
5. Rename the resulting folder to exactly: **psid-shelf-app** — no spaces, no "-main" suffix.
 - Mac: right-click the folder → Rename.
 - Windows: right-click → Rename.

Folder renamed to psid-shelf-app in Downloads

✗ Do not skip the rename step. A folder named "psid-shelf-app-main" will cause "No such file or directory" errors later.

PART 2

Check Python and install required packages

Step 1 — Open a terminal

- **Mac:** press Cmd+Space, type "Terminal," press Enter.
- **Windows:** press the Windows key, type "PowerShell," press Enter.
- **Linux Mint:** press Ctrl+Alt+T.

Terminal window open and ready — blank prompt visible

Step 2 — Check that Python is installed

Type the following and press Enter:

```
python3 --version
```

Terminal showing python3 --version and the Python version number in response

- If you see something like **Python 3.11.4** — Python is installed. Continue to Step 3.
- If you see "command not found" — install Python first:
 - Mac: install Homebrew from brew.sh, then run **brew install python3**.
 - Windows: download from python.org/downloads. During install, tick "Add Python to PATH" — this is critical.
 - Linux: run **sudo apt update && sudo apt install python3**

Step 3 — Install the required packages

The data preparation script needs two Python packages that are not installed by default. Type the command for your operating system and press Enter:

Mac / Linux

```
pip3 install pandas pyreadstat --break-system-packages
```

Windows

```
pip install pandas pyreadstat
```

pip install running — lines of download activity scrolling in terminal

Wait for it to finish. You will see several lines of download activity — this is normal. When you see **"Successfully installed"** near the bottom, you are ready.

Successfully installed message at bottom of terminal output

Confirm it worked:

Mac / Linux

```
python3 -c "import pandas; import pyreadstat; print('Ready')"
```

Windows

```
python -c "import pandas; import pyreadstat; print('Ready')"
```

You should see **Ready** printed. If you see an error, do not proceed — paste the error into Claude or a search engine to diagnose.

Run the app in Preview mode

Let's confirm the app works before dealing with the data.

Step 1 — Open a new terminal window

Open a fresh terminal (same method as Part 2, Step 1).

Step 2 — Navigate to the app folder

You need to point the terminal at the `psid-shelf-app` folder you created in Part 1.

Easiest method — drag and drop (Mac and Linux)

1. Type `cd` followed by a single space. Do not press Enter yet.
2. Open Finder (Mac) or your file manager (Linux).
3. Find the `psid-shelf-app` folder in Downloads.
4. Drag that folder onto the terminal window. The full path fills in automatically.
5. Press Enter.

Dragging the psid-shelf-app folder onto the terminal window — path appears automatically

Type the path manually (all platforms)

Mac / Linux

```
cd ~/Downloads/psid-shelf-app
```

Windows

```
cd %USERPROFILE%\Downloads\psid-shelf-app
```

Windows shortcut

In File Explorer, navigate to the `psid-shelf-app` folder. Shift+right-click inside it and choose "Open PowerShell window here" or "Open in Terminal." No typing required.

Confirm you are in the right place:

```
ls
```

You should see `index.html` in the list. If you do not, you are in the wrong folder — repeat Step 2.

ls output showing index.html and other app files

Step 3 — Start the local server

Mac / Linux

```
python3 -m http.server 8000
```

Windows

```
python -m http.server 8000
```

You will see: **Serving HTTP on 0.0.0.0 port 8000 ...**

Terminal showing the Serving HTTP message

✗ Leave this terminal window open. It is the server. Closing it stops the app.

Step 4 — Open the app

Open your web browser and go to:

```
http://localhost:8000
```

PSID Studio loading in the browser — gold PREVIEW · 10-ROW SAMPLE badge visible in corner

You should see PSID Studio with a gold **PREVIEW · 10-ROW SAMPLE** badge. This is correct. You can now search all 34 topics and preview 10 rows of each.

To stop the app when you are done: click back into the terminal and press Ctrl+C.

PART 4

Download the SHELF data from ICPSR

⚠ The download is about 2.4 GB. Make sure you have enough disk space and a stable internet connection. It may take 10–30 minutes.

1. Go to openicpsr.org and log in.
2. In the search box, type or paste this DOI: **10.3886/E194322** and press Enter.

ICPSR search box with DOI 10.3886/E194322 entered

3. Click on the result: **PSID-SHELF, 1968–2021**.

ICPSR search result showing PSID-SHELF entry

4. Make sure you are on **Version 2.1** — check the version number on the page.

Version 2.1 label on the ICPSR dataset page

5. Click the **Download** button.

Download button on the ICPSR page

6. Your browser saves **ICPSR_194322-V2.1.zip** to your Downloads folder. Wait for it to finish.

Unzip the download — this is a two-step process

1. Find **ICPSR_194322-V2.1.zip** in your Downloads folder and unzip it (double-click on Mac, right-click → Extract All on Windows, right-click → Extract Here on Linux).

This creates a folder called **ICPSR_194322-V2.1**. Inside you will see four items:

```
Construction_Files/  
Documentation/  
PSIDSHELF_1968_2021_LONG_7.9_GB.zip  
PSIDSHELF_1968_2021_WIDE_2.4_GB.zip
```

Contents of ICPSR_194322-V2.1 folder showing the four items

2. **You want only the WIDE zip.** Ignore everything else. Unzip **PSIDSHELF_1968_2021_WIDE_2.4_GB.zip** the same way.

⚠ Do not unzip the LONG zip or the Construction_Files zip. You only need the WIDE file.

3. This produces one file: **PSIDSHELF_1968_2021_WIDE.dta** (~2.2 GB). Leave it inside the **ICPSR_194322-V2.1** folder.

The .dta file sitting inside the ICPSR_194322-V2.1 folder

Prepare the data

This step reads `PSIDSHelf_1968_2021_WIDE.dta` and splits it into 34 smaller CSV files, one per topic. The app reads these smaller files.

✗ Do not close the terminal while this is running. The script is reading a 2.2 GB file and it takes several minutes.

Step 1 — Navigate to the data folder

Open a terminal and navigate to the `ICPSR_194322-V2.1` folder — the one containing the .dta file.

Easiest method — drag and drop (Mac and Linux)

1. Type `cd` followed by a single space. Do not press Enter yet.
2. Find the `ICPSR_194322-V2.1` folder in your file manager.
3. Drag it onto the terminal window. The path fills in automatically.
4. Press Enter.

Dragging ICPSR_194322-V2.1 folder onto the terminal window

Type the path manually

Mac / Linux

```
cd ~/Downloads/ICPSR_194322-V2.1
```

Windows

```
cd %USERPROFILE%\Downloads\ICPSR_194322-V2.1
```

Step 2 — Run the script

Mac / Linux

```
python3 ~/Downloads/psid-shelf-app/prepare_data.py
```

Windows

```
python %USERPROFILE%\Downloads\psid-shelf-app\prepare_data.py
```

Terminal with the prepare_data.py command typed and ready to run

Press Enter. The script reads the .dta file (this takes a few minutes), reports how its column mapping lines up with what's in the file, then writes each topic CSV with its column count and file size:

```

✓ Loaded: 84,121 rows × 7,525 columns
Columns expected by the mapping: 6,588
Columns present in the .dta:      7,525
Missing (in mapping, not in dta): 0
Extra (in dta, not in mapping):  937

✓ Every expected column is present.
Writing 34 topic CSVs to shelf_csvs/

✓ covid                18 cols      3.1 MB
✓ demographics         92 cols     16.2 MB
✓ depression           93 cols     10.6 MB
...
✓ health_hospitalization 35 cols      4.4 MB

✓ Done. 34 files written to shelf_csvs/

```

Terminal showing the completion message — all 34 topics written

The "Extra" column count is normal — those are .dta columns not used by any SHELF topic. As long as "Missing" reads 0, every topic the app needs was found.

When it finishes, a folder called `shelf_csvs` appears inside `ICPSR_194322-V2.1`, containing 34 CSV files.

△ If the command fails with "No such file or directory" for `prepare_data.py`, the app folder name may differ. Run `ls ~/Downloads | grep psid` (Mac/Linux) to see the exact name, then adjust the path in the command above.

PART 6

Load your data into the app

1. Make sure the app is running (Part 3). If you stopped it, repeat Part 3 Steps 2–4 to restart it.
2. In the app, find the yellow "Load my CSVs" strip near the top and click it.

Yellow Load my CSVs strip near the top of the app

3. A file picker opens. Navigate to: Downloads → ICPSR_194322-V2.1 → shelf_csvs

File picker open showing the shelf_csvs folder with 34 CSV files

4. Select all 34 files at once: press Cmd+A on Mac or Ctrl+A on Windows/Linux, then click Open.
5. The badge changes from gold `PREVIEW · 10-ROW SAMPLE` to green `YOUR DATA · 100-ROW PREVIEW`.

Green YOUR DATA · 100-ROW PREVIEW badge in the app corner

You will also see: "Loaded 34 topics. Showing first 100 rows."



You're done. PSID Studio is running on your full SHELF data.

Troubleshooting

What you see	What to do
Terminal shows "Killed"	Your computer ran out of RAM. The script needs at least 8 GB free. Close all other apps and try again. If it still fails, run it on a machine with more memory and copy the shelf_csvs folder over.
"python3: command not found"	Python is not installed. See Part 2, Step 2. On Windows, try <code>python</code> instead of <code>python3</code> .
"File not found: PSIDSHELF_1968_2021_WIDE.dta"	You ran the script from the wrong folder. Make sure you navigated to <code>ICPSR_194322-V2.1</code> first (Part 5, Step 1), not the app folder.
"No module named 'pyreadstat'" or "No module named 'pandas'"	Run the install command from Part 2, Step 3 again and retry.
Browser shows a folder listing instead of the app	You started the server in the wrong folder. The terminal must be inside <code>psid-shelf-app</code> (the folder with <code>index.html</code>). Repeat Part 3, Step 2.
Badge still says PREVIEW after loading CSVs	Check: (a) you opened the <code>shelf_csvs</code> folder, not <code>ICPSR_194322-V2.1</code> itself; (b) you selected all 34 files at once; (c) filenames are plain — <code>covid.csv</code> , not <code>psid_shelf_covid.csv</code> .
Only some topics loaded	A file is missing. A complete load shows ~6,585 variables. Check that <code>prepare_data.py</code> finished with: ✓ All 34 topics written to <code>shelf_csvs/</code>
Closing the terminal stopped the app	Expected — the terminal is the server. Open a new terminal and repeat Part 3, Steps 2–4 to restart.
"Address already in use"	Another server is running on port 8000. Use a different port: <code>python3 -m http.server 8080</code> then open <code>http://localhost:8080</code> .
"No such file or directory" for <code>prepare_data.py</code>	The app folder name differs. Run <code>ls ~/Downloads grep psid</code> (Mac/Linux) or <code>dir %USERPROFILE%\Downloads</code> (Windows) to find the exact name, then adjust the command.

FAQ

Do I need to download data just to try the app?

No. The online version at psid-shelf-app.netlify.app works immediately with no download.

Why isn't the SHELF data included in the download?

The SHELF data is licensed through ICPSR and cannot be redistributed. Each user downloads it under their own ICPSR account.

Do I need Jupyter, Stata, or any other software?

`prepare_data.py` is a plain Python script. Just run it in the terminal as shown in Part 5. No Jupyter, no Stata, nothing else needed.

Does the online demo have all 34 topics now?

Yes. The online demo (PSID Studio Lite) and the local version you are installing here both cover all 34 topics. The only difference is how much data you see per topic: 10 baked-in preview rows online versus 100 of your own rows once you load your local CSVs in Part 6.

Is the PSID Crosswalk tab working?

Not yet. It is a "Coming Soon" stub. Mapping SHELF variable names back to raw PSID variable names is planned for a future release.

Can I use a different port than 8000?

Yes: `python3 -m http.server 8080` then open `http://localhost:8080`.

The data and the app are on different machines. What do I do?

Run `prepare_data.py` on the machine where the .dta file lives. Then copy the `shelf_csvs` folder to the machine running the app. Load the CSVs from wherever you copied them.

Citation & license

If you use PSID-SHELF data in published work, please cite:

*Pfeffer, Daumler & Friedman. **PSID-SHELF, 1968–2021**. DOI: 10.3886/E194322*

The app code is provided as-is. The SHELF dataset is governed by ICPSR terms of use — review them at OpenICPSR before redistributing anything.